Assignment 3- Linear and Logistic Regression

Air quality is a critical factor that directly impacts the health and well-being of individuals and the environment. Poor air quality can lead to various respiratory and cardiovascular issues, as well as environmental degradation. Machine learning, with its ability to process large volumes of data and identify complex patterns, can play a vital role in air quality monitoring and management. By analyzing data from sensors, satellite imagery, weather patterns, and historical records, machine learning algorithms can accurately predict and detect air pollution levels, identify pollution sources, and help in developing effective mitigation strategies.

Because the air helps the skin expel moisture, humans are particularly sensitive to humidity. Your body tries to stay cool and maintain its present temperature by sweating. Sweat won't evaporate into the air if there is a 100% relative humidity in the air. As a result, when the relative humidity is high, we experience heat much more intensely than the actual temperature. When the relative humidity is low, our sweat evaporates quickly, cooling us off, making us feel considerably colder than the actual temperature. When the relative humidity is zero percent and the air temperature is 75 degrees Fahrenheit (24 degrees Celsius), for instance, our bodies perceive the air temperature as 69 degrees Fahrenheit (21 degrees C). When the relative humidity is 100% and the air temperature is 75°F (24°C), we perceive a temperature of 80°F (27°C).

The airquality.csv dataset shared on Canvas contains the results from a gas multisensor device that was placed in a city in Italy. Averages of the hourly replies are noted together with references to gas concentrations from a licensed analyzer. You can get more information about the data here.

Attribute Information:

Variable Name	Description
PT08.S1(CO)	PT08.S1 (tin oxide) hourly averaged sensor response (nominally CO targeted)
NMHC(GT)	True hourly averaged overall Non Metanic HydroCarbons concentration in microg/m^3 (reference analyzer)
PT08.S2(NMHC)	PT08.S2 (titania) hourly averaged sensor response (nominally NMHC targeted)
NOx(GT)	True hourly averaged NOx concentration in ppb (reference analyzer)
PT08.S3(NOx)	PT08.S3 (tungsten oxide) hourly averaged sensor response (nominally NOx targeted)
NO2(GT)	True hourly averaged NO2 concentration in microg/m^3 (reference analyzer)
PT08.S4(NO2)	PT08.S4 (tungsten oxide) hourly averaged sensor response (nominally NO2 targeted)
PT08.S5(O3)	PT08.S5 (indium oxide) hourly averaged sensor response (nominally O3 targeted)
T	Temperature in °C
RH	Relative Humidity (%)

Part 1- Linear Regression:

Design a Linear Regression Model to predict RH from the given variables. Explain the R-squared. Also, plot 2 variables of your choice and fit a linear model on them.

Part 2- Logistics Regression:

For this part, we need a categorical target but as you see our target is numerical. We use following strategy to convert the numerical variable to categorical:

- RH< 492   RH_categorical=0
- RH=> 492   RH_categorical=1

You will add this new column and drop the column named RH for this part.

Now, design a Logistic Regression Model to predict RH_categorical from the given variables.

You are free to use either R or Python for this assignment, but the submission should be in HTML format. Note that if the variable is in the wrong data type. Make sure to convert the target variable into the appropriate type.

This assignment contains two files:

1) Instructions (this file)

2) airquality.csv